

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

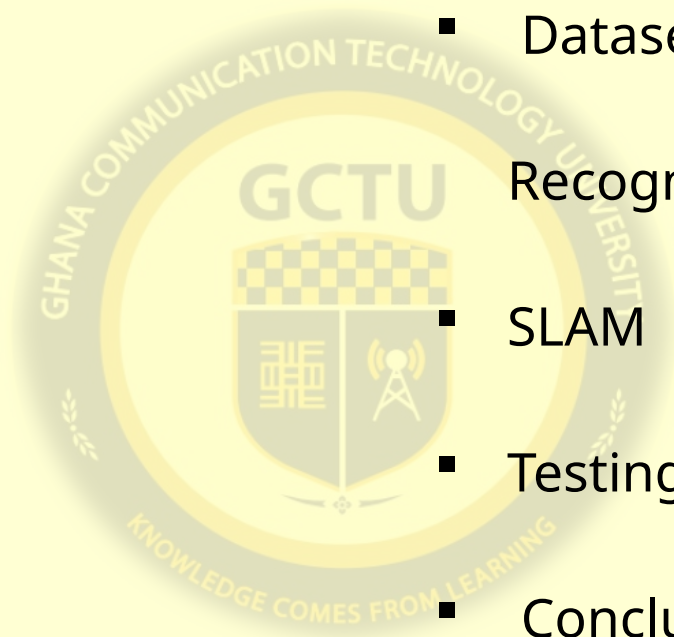
TITLE:
**DEVELOPMENT AND IMPLEMENTATION OF A ROBOTIC
APPLICATION FOR HUMAN INTENTION RECOGNITION
USING MOTION AND HAND GESTURE**

A Project Work Topic Proposal in Partial Fulfillment of the
requirements for BSc. Computer Engineering

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PRESENTATION OUTLINE

- Introduction
- Problem Statement
- Objectives of Study
- Significance of Study
- Research Gaps
- Literature Review
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- Dataset Preparation and Gesture Recognition
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INTRODUCTION

- **Human-Robot Interaction (HRI)** enables robots to understand and respond to human actions.
- **Gestures and body movements** provide natural, contactless control.
- **Vision-based systems** can require high computing power..
- **Edge computing** supports real-time processing on low-power robots.
- This project combines **intention recognition, environmental awareness, and navigation.**



Figure 1: Autonomous Mobile Robot Performing Navigation Tasks



Figure 2: Human Interaction with an Autonomous Mobile Robot Interface

Problem Statement

- **Existing systems** often separate gesture recognition and navigation.
- **High computing demands** limit edge deployment.
- **Cloud dependence** affects real-time operation.
- **Data mismatch** can reduce recognition accuracy.
- **Limited environmental awareness** creates safety risks.



OBJECTIVES

General Objective

To design, implement and evaluate an edge-based real-time human intention recognition system integrating hand gestures and autonomous navigation using ROS 2..

Specific Objectives

- Develop a real-time landmark-based MLP gesture classifier.
- Train a reliable classifier using a balanced custom dataset.
- Implement multi-node communication and centralized decision-making.
- Integrate SLAM and navigation for gesture-directed movement.
- Measure accuracy, latency and hardware performance.

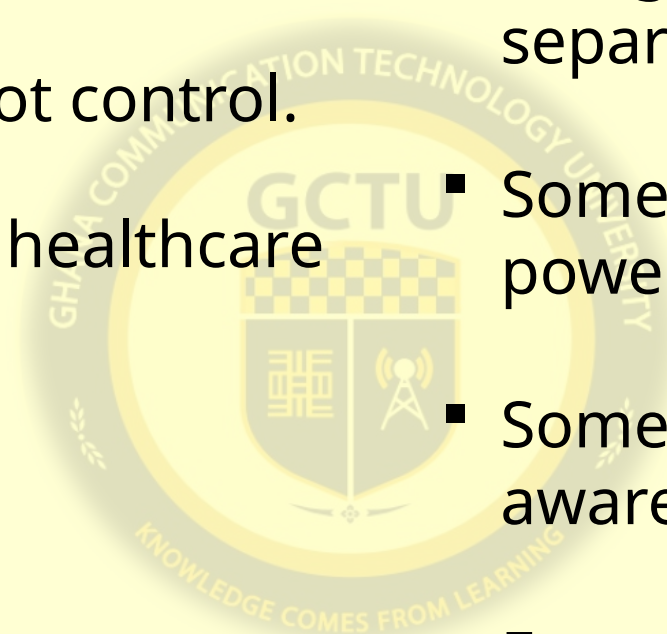


SIGNIFICANCE OF THE STUDY

- Reduces dependence on cloud computing.
- Enables contactless robot control.
- Supports industrial and healthcare applications.
- Improves human-robot interaction.
- Enables AI processing on edge devices

RESEARCH GAPS

- Existing systems often treat recognition and navigation separately.
- Some systems depend on cloud or powerful GPUs.
- Some lack environmental awareness.
- Few combine intention recognition with navigation.
- his study integrates these functions on an edge robot



LITERATURE REVIEW					
Ref / Year	Author(s)	Topic	Objectives	Methodology	Identified Gaps
[1] 2022	Tsitos et al.	Real-time Feasibility of a Human Intention Method Evaluated Through a Competitive Human-Robot Reaching Game	Predict human intention in real time. ing a single RGB-D camera	OpenPose, RGB-D camera, ML models, ROS, UR3 robot.	Limited integration of multimodal human intention recognition.
[2] 2022	Mahmud et al	3D Gesture Recognition and Adaptation for Human-Robot Interaction	Recognize 3D gestures for robot control.	Kinect, HMM, SVM, CNN, robot navigation.	Depends on Kinect, reducing portability and affordability. Requires specialized 3D sensing hardware. Limited applicability to low-cost edge

METHODOLOGY

System Architecture



Hardware & Software Setup



Dataset Collection and Preprocessing



Gesture Recognition Pipeline



Brain Node Decision Making



ROS2 Integration



SLAM & Navigation



SYSTEM ARCHITECTURE

Figure 3: Hardware Architecture and Signal distribution

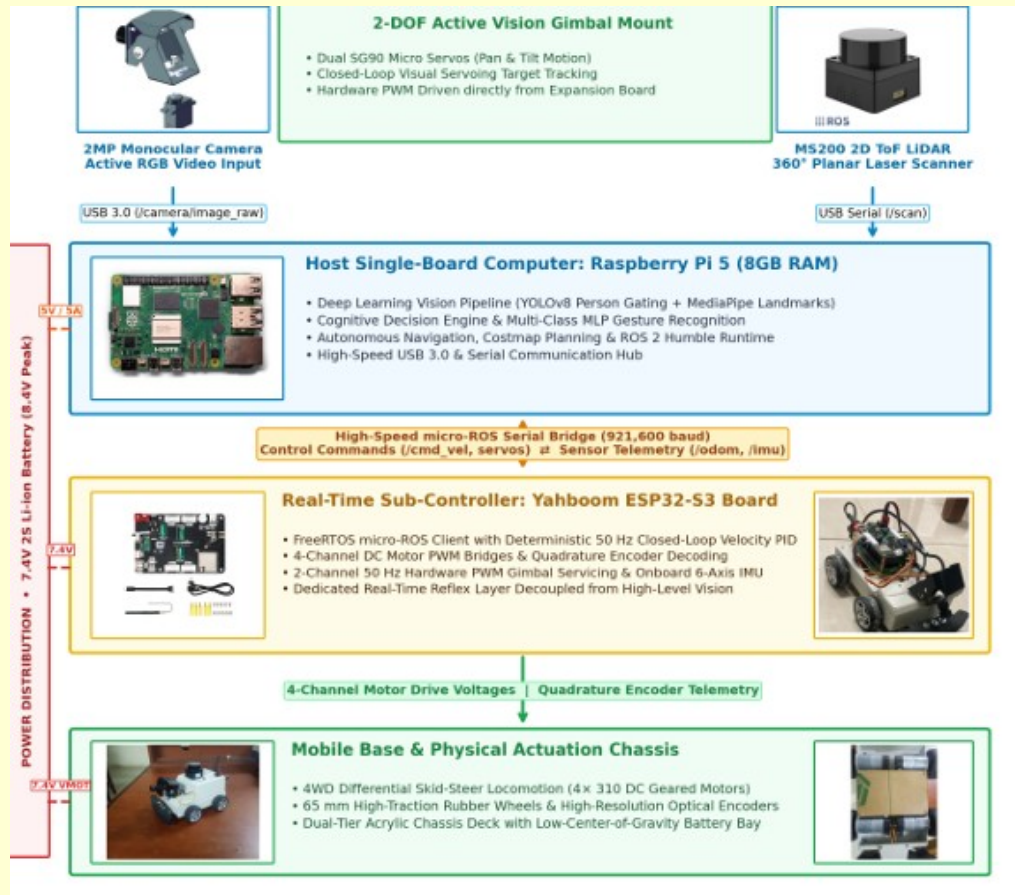
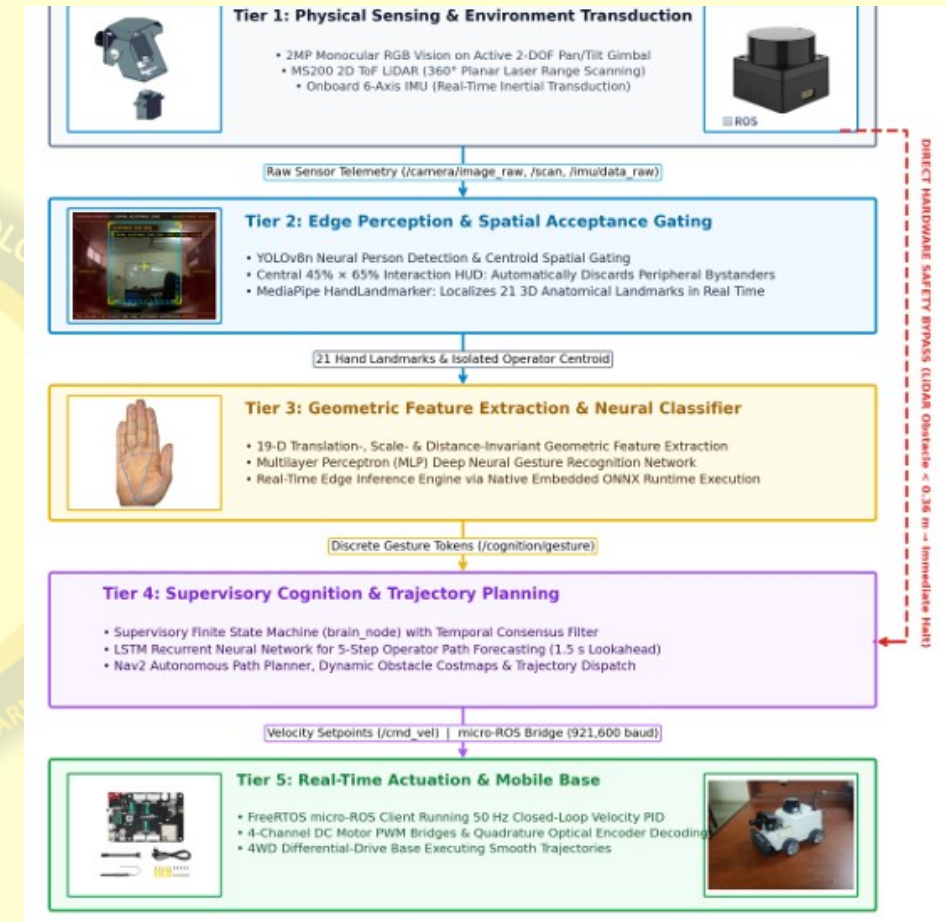


Figure 4: System Architecture



HARDWARE COMPONENTS



Figure 5: Raspberry Pi 5 :Raspberry Pi 5: Main computer that processes **AI, camera, LiDAR, and SLAM data** directly on the robot..



Figure 6: USB Camera : Captures gestures and keeps the user centered for tracking.



Figure 7: M200 LiDAR Sensor The LiDAR detects obstacles, supports SLAM mapping, and stops the robot within the **0.36 m safety zone.**"

HARDWARE COMPONENTS



7.4V power supply

Figure 8: Power Supply/Battery :Uses a 7.4V rechargeable battery with separate power for the motors and Raspberry Pi..

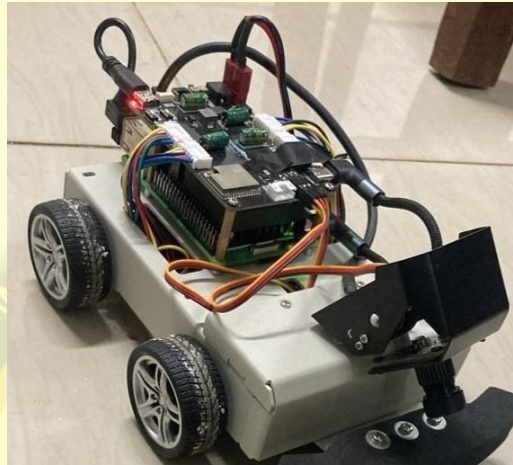


Figure 9:Uses a Yahboom MicroROS-Pi5 differential-drive robot with four geared motors and wheel encoders .

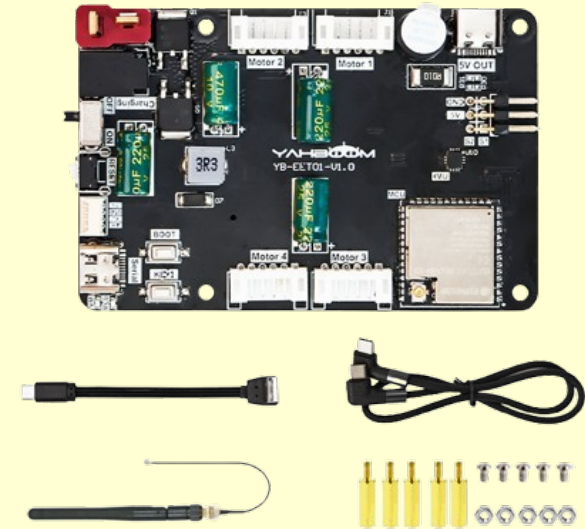


Figure 10: The Micro-ROS board controls the motors and sensors. It uses an ESP32-S3 and communicates with the Raspberry Pi."

DATASET PREPARATION AND GESTURE RECOGNITION PIPELINE

- **6,000 samples** collected from the robot's own camera.
- **6 gesture classes:** Stop, Go, Left, Right, Back, Follow.
- **Balanced dataset:** 1,000 samples per gesture.
- **MediaPipe** extracts 21 hand landmarks
- Landmarks are converted into **19 normalized geometric features**
- **MLP classifier** recognizes the six gestures..

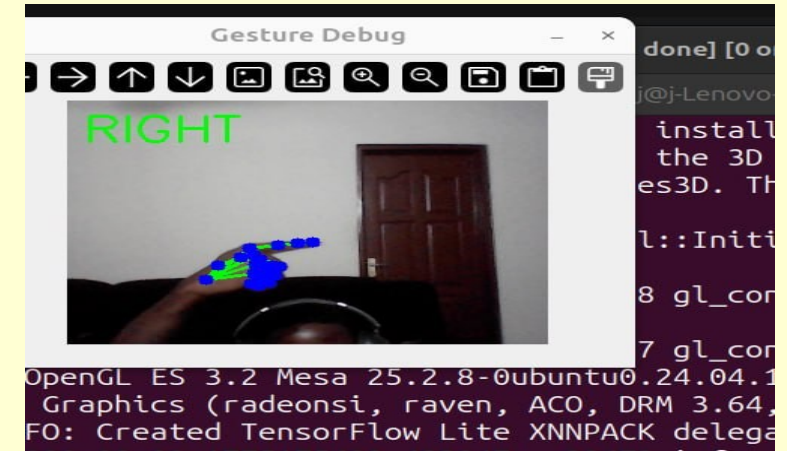


Figure 11: Live System Demonstration: The detected gesture is classified and displayed in real time.

HUMAN MOVEMENT PREDICTION (LSTM)

- Uses **MediaPipe Pose landmarks** to track human movement.
- LSTM predicts the operator's next movement direction.
- Uses **past movement patterns** to anticipate future motion.
- Helps the robot **respond proactively instead of reactively**.
- Achieved **25.8 px ADE** and **32.4 px FDE**.

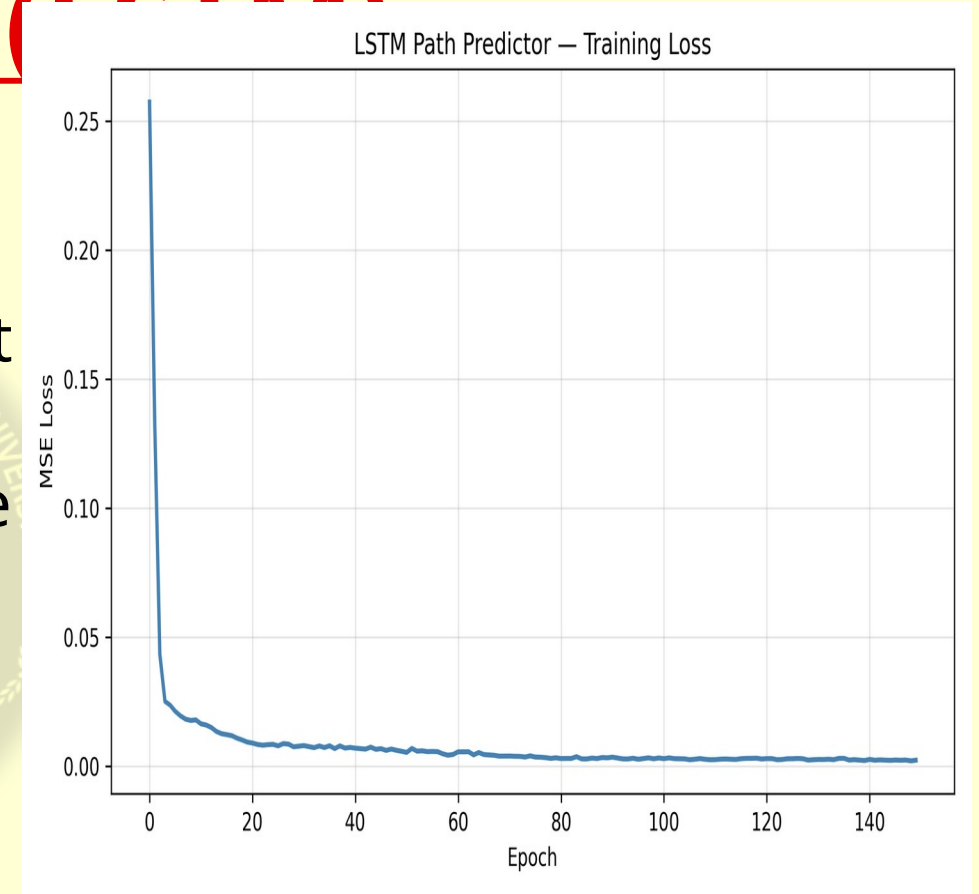


Figure 13: LSTM Path Predictor Training and Validation Loss Convergence

SLAM MAPPING AND LOCALISATION

- **SLAM Toolbox** creates a 2D map using LiDAR and wheel odometry.
- The environment is represented as an **occupancy grid**.
- **Nav2** enables point-to-point autonomous navigation.
- **AMCL** localizes the robot on the generated map.
- **Costmaps** help the robot avoid unexpected obstacles



Figure 13:Initial Run:
"Hourglass" Rotational Drift
Defect.

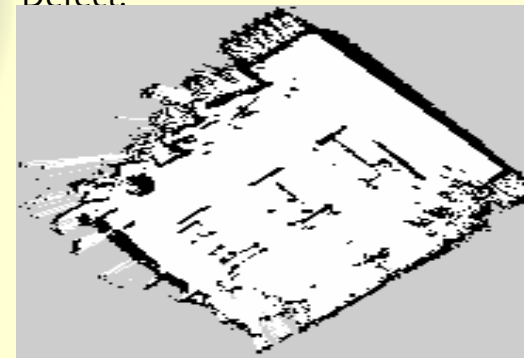


Figure 14:Final Run (3rd
Attempt): Clean Metric Map

TESTING AND RESULTS

Gesture Recognition Physical Trial Results

Gesture	Total Trials	Successful Trials	Observed Accuracy (%)
Stop	30	27	90.0%
Go	30	29	96.7%
Follow	30	28	93.3%
Left	30	30	100.0%
Right	30	30	100.0%
Back	30	30	100.0%
Aggregate	180	174	96.67%

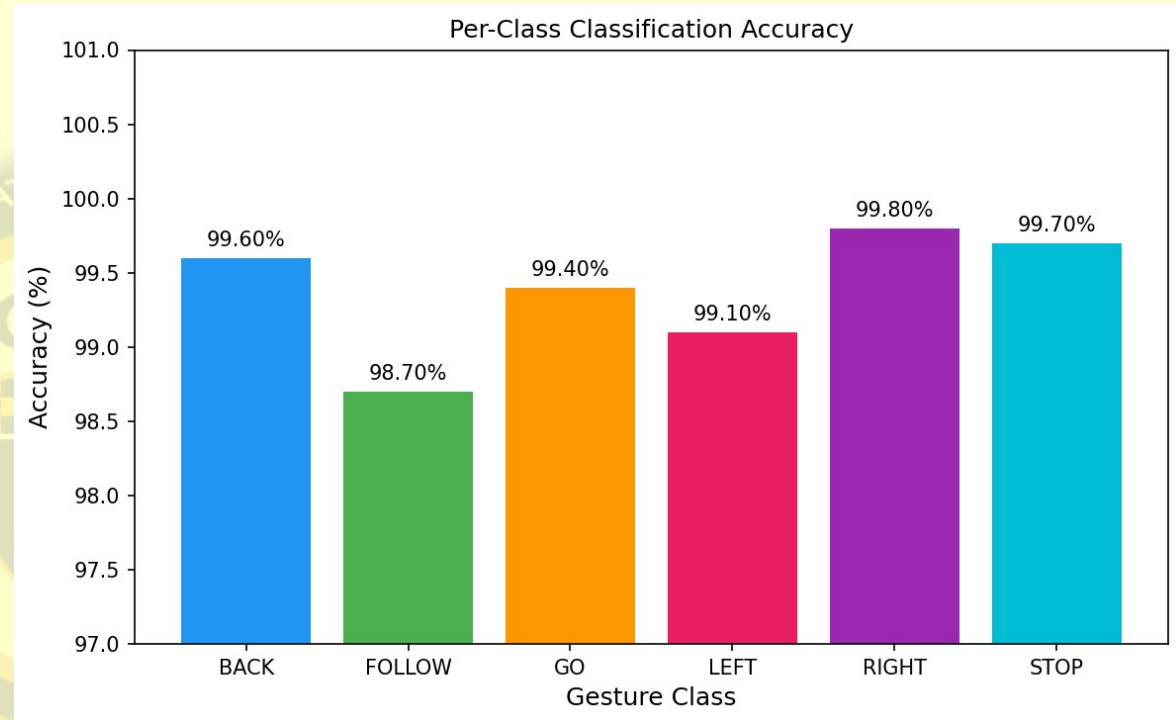


Figure 15: Per-Class Classification Accuracy of the Trained MLP Gesture Model



TESTING AND RESULTS

Movement Recognition Test Results

Movement Class	Observed Kinematic Behavior	Result
Approaching	Rapid expansion of bounding box area with stable center	Successfully Recognized
Moving Away	Continuous contraction of bounding box area with stable center	Successfully Recognized
Moving Left	Persistent leftward centroid translation across frames	Successfully Recognized
Moving Right	Persistent rightward centroid translation across frames	Successfully Recognized
Stationary	Centroid displacement within noise threshold (± 4 px)	Successfully Recognized



CONCLUSION AND FUTURE WORKS

Conclusion

- 99.38% gesture recognition accuracy.
- 96.67% physical trial accuracy.
- Successful human movement prediction.
- ROS 2 integration achieved.
- SLAM and Nav2 validated
- Real-time edge operation achieved
- Safe obstacle detection implemented

Future works

- Multi-waypoint autonomous navigation.
- Proactive human-aware navigation.
- Follow-to-Map exploration.
- NPU acceleration for AI tasks.
- Improved gimbal-chassis control
- Multi-sensor safety detection
- Larger and more diverse testing.



REFERENCES

- [1] D. Tsitos *et al.*, "Real-Time Feasibility of a Human Intention Prediction Method Evaluated Through a Competitive Human–Robot Reaching Game," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10034–10041, Oct. 2022.
- [2] J. A. Mahmud, B. C. Das, J. Shin, K. M. Hasib, R. Sadik, and M. F. Mridha, "3D Gesture Recognition and Adaptation for Human–Robot Interaction," *IEEE Access*, vol. 10, pp. 116485–116507, 2022.
- [3] G. Yang, S. Wang, H. Okamura, Y. Ueda, T. Yasui, T. Yamada, Y. Miyazawa, S. Yoshida, Y. Inada, S. Ino, K. Okuhata, and Y. Mizobuchi, "Safe and Efficient Motion Planning for Material Transportation Robots Considering Intention Prediction of Obstacles," in *Proc. IEEE Int. Conf. Mechatronics Autom. (ICMA)*, Harbin, China, Aug. 2023, pp. 1179–1184.



Questions And Discussion
Thank you



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..Knowledge Comes From Learning